

Organic matter accumulation in reclaimed soils under spruce, poplar and grass in the Alberta Oil Sands

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Abstract Fundamental to the success of forest restoration following major disturbances such as mining is development of a functioning soil, including the amount, properties and rates of accumulation of soil organic matter (SOM). SOM enters mineral soil through leaching of dissolved organic matter from the forest floor; macrofaunal processing of above-ground litter and mixing with soil (bioturbation); and direct deposition of root litter and rhizodeposits. Our study focused on how SOM accumulation in reclaimed mineral soils was affected following re-vegetation with three vegetation treatments: deciduous trees (*Populus tremuloides* Mitchx. and *Populus balsamifera* L.), spruce trees (*Picea glauca* (Moench) Voss.) and grasses (*Festuca* sp., *Bromus inermis*), compared to the natural boreal forest. Seventeen sites were studied: 4 reclaimed Deciduous, 5 reclaimed Spruce, 4 reclaimed Grass, and 4 Natural forest. Concentrations of soil organic matter in the upper 30 cm of soil were highest at the Deciduous sites and lowest at Natural sites. SOM concentrations in the top 10 cm of soil were elevated relative to 10–30 cm soil at the Deciduous and Grass sites. The elevated SOM in the upper soil at Deciduous and Grass sites was associated with greater macrofaunal activity and higher root biomass at these sites. The higher macrofaunal activity in the surface organic layer and greater amounts of faunal fecal material in the uppermost cm of soil at Deciduous and Grass sites indicate greater enrichment of upper soil by macrofauna at these sites. SOM concentrations were significantly positively related to root abundance at Grass sites (indicating a contribution of roots to SOM), and a similar trend was apparent at the Deciduous sites. The elevated organic matter concentrations in the upper 10 cm of soil at the Deciduous and Grass sites suggest that planting of aspen following reclamation would hasten C sequestration into soil organic matter.

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Introduction

Successful restoration of forest ecosystems requires the development of fundamental physical, chemical and biological properties of the soil. A critical facet of soil reclamation is restoration of soil organic matter (SOM), in amount, properties and rates of accumulation. The positive effects of organic matter on soil fertility include: buffering pH, reducing bulk density, and raising cation-exchange capacity, water-holding capacity and nutrient availability (Hudson 1994; Brady and Weil 2008). SOM is also an important carbon sink, as CO₂ captured from the atmosphere through photosynthesis is stabilized to various degrees as SOM (Harrison et al. 1994). Organic matter enters the soil through three principal mechanisms: leaching of dissolved organic matter (DOM) from the forest floor into the mineral soil (Sanderman and Amundson 2003); macrofaunal mixing of above-ground litter from the forest floor into mineral soil layers (bioturbation); and direct deposition of root litter and rhizodeposits into mineral soil. Each of these mechanisms is influenced by the vegetation present at the site: the composition and amount of plant litter influences the composition and concentration of DOM entering the soil (Kalbitz et al. 2000; Marschner and Kalbitz 2003) and the nature of the macrofaunal communities responsible for bioturbation (Lavelle et al. 1997; Frouz et al. 2001, 2006). Likewise, roots of different plant species have different properties, turnover rates, exudate profiles and fungal associations, each of which influences the quantity and nature of SOM inputs to the soil (Canadell et al. 1996; Finér et al. 1997; Grayston et al. 2000; Treseder and Allen 2000; Chen and Brassard 2013).

Accumulation of soil organic matter in reclaimed mine soils tends to be concentrated in upper soil layers (Rumpel and Hüttl 1999; Sourkova et al. 2005; George et al. 2010; Lin et al. 2011), and is usually rapid during the first decade or two, and then slows (Schwenke et al. 2000; Bendfeldt et al. 2001; Sourkova et al. 2005; Ussiri and Lal 2005; Amichev et al. 2008; Banning et al. 2008; Chaudhuri et al. 2013). The nature of the vegetation on reclaimed sites influences the rate of accumulation, depth distribution and properties of soil organic matter. Slower soil C accumulation has been reported in mine soils under forest cover than under grass cover (Akala and Lal 2001) or pasture (Sperow 2006). Among tree species, higher rates of soil C accumulation have been reported under broadleaf species than under conifers (Frouz et al. 2009), with particularly high rates under species with associative N-fixers (Chiti et al. 2007; Chodak and Niklinska 2010). In a meta-analysis, Vindušková and Frouz (2013) found that mean soil organic carbon (SOC) accumulation rates were significantly lower in coniferous forests than in grasslands or in deciduous forests, but did not differ between grasslands and deciduous forests.

In the Athabasca Oil Sands Region (AOSR) of Alberta, Canada, open-pit mining is employed where the oil sands deposit is within approximately 100 m of the surface. Suitable surface soils from upland boreal forests and lowland peat bogs and fens are salvaged for use in reclamation, followed by removal of non-ore horizons (typically referred to as “overburden”) to expose the oil-sand ore deposit. Mining, extraction and refining operations result in a number of substrates and landforms that require reclamation. Approximately half of the total disturbed area is peat-forming wetlands (Vitt et al. 2000), and this peat, along with a portion of the underlying mineral parent material (if it has suitable soil physical and chemical properties), is used to create a peat-mineral mix (PMM). PMM is used as coversoil in reclamation and may be underlain by mineral subsoil material salvaged

for use in reclamation (depending on the landform or substrate being reclaimed). PMM coversoil depths generally range from 20 to 30 cm. Barley (*Hordeum vulgare*) is seeded as a cover crop, and native tree species (*Populus tremuloides*, *Picea glauca*) are planted during the same year or the following growing season after barley seeding.

Soil organic matter content has been demonstrated to be critical for forest restoration on reclaimed oil sands, with growth rates of trees in reconstructed oil sand soils positively correlated with SOM content up to 8% OM (dry weight basis; Farnden et al. 2013). Although the coversoil contains additional organic matter in the form of peat, this material, like other organic amendments (e.g. biosolids), may not provide all of the benefits associated with naturally-formed soil organic matter (Vetterlein and Hüttl 1999). Formation of new soil organic matter is, therefore, an important process in these soils, as in other reclaimed mine soils. In this study we measure amounts of soil organic matter in PMM under three common revegetation treatments in the AOSR: deciduous (*Populus tremuloides* Mitchx. and *Populus balsamifera* ssp. *balsamifera*), white spruce (*Picea glauca* (Moench) Voss.) and grass (*Festuca*, *Bromus* and *Agropyron*), 20–36 years after planting and/or seeding. We looked for evidence of formation of new soil organic matter by (1) examining the distribution of SOM in the upper 30 cm of the soil profile, and (2) quantifying abundances of faunal fecal matter and roots in the soils in each treatment and relating these to concentrations of SOM. We hypothesized that grass sites would have the greatest abundance of roots and high concentrations of SOM, deciduous sites will have the most bioturbation and high concentration of SOM in the upper 1 cm of soil, and that coniferous sites would have the lowest abundance of SOM, roots and bioturbation. Our primary objective is to determine which of the vegetation treatments is most effective in building soil organic matter levels in the first few decades following reclamation. For purposes of comparison with a natural baseline, we include similar measurements of SOM in natural boreal forest soils in the region.

Methodology

Study sites

The study sites are located in the Athabasca Oil Sands region north of Fort McMurray, in northeastern Alberta, Canada. The climate is characterized by relatively short cool summers and long cold winters. Mean monthly air temperatures range from -18.8°C in January to 16.8°C in July, with a mean annual temperature of 0.7°C (Environment Canada 2013). Mean annual precipitation is 456 mm, of which 342 mm occurs as rainfall during the growing season (Environment Canada 2013). Upland vegetation in the region is classified as Boreal Mixedwood and dominated by aspen (*Populus tremuloides* Mitchx.) and white spruce (*Picea glauca* (Moench) Voss.), with jack pine (*Pinus banksiana* Lamb.) occupying drier sites (Beckingham et al. 1996). Soils of the region are derived primarily from glaciofluvial and glacial till parent materials, the most common upland soil types found in the region are Orthic Gray Luvisols and Eutric Brunisols (Beckingham et al. 1996).

The following criteria were used to select reclaimed sites: upland, mesic, peat-mineral-mix coversoil, and > 20 years since reclamation. Sites meeting these criteria included four deciduous sites, five spruce sites and four grassland sites. Following Beckingham et al. (1996), a site was considered deciduous if the canopy cover was at least 80% deciduous,

while spruce sites had to have a canopy of at least 80% spruce. Sites were considered grass if tree cover was less than 10% and shrub cover was less than 20%. Deciduous and spruce sites were all between 20 and 28 years since planting. Grass sites ranged from 27 to 36 years and had the same soil treatments as treed sites, but had not been planted with trees or shrubs in order to facilitate access to communication towers or to retain sight-lines.

Natural forests of a similar age (after a stand-replacing wildfire) were used as natural analogues. Historical fire data were used to find natural stands of a similar age, and regional soil-survey data were used to ensure that the natural forest sites were of similar soil texture and parent material to that which would have been found at the reclamation sites prior to disturbance. Four sites which were of a similar moisture and nutrient regime to the reclaimed sites were selected, which corresponded to ecosite types “b (blueberry)” “c (Labrador tea mesic)” or “d (low-bush cranberry)” (Beckingham et al. 1996). Natural sites were not selected based on tree species, as early successional pathways in this region do not favour spruce. The nearest accessible natural sites were 40–90 km south-west of Fort McMurray and ranged in age from 10 to 40 years.

Measurements

A 10 × 10-m plot was established at each site, within which three 1 × 1-m sub-plots were randomly selected for study, excluding any that contained the base of a tree or shrub. Cover of tree and shrub species was estimated at the plot level. In each 100-m² plot, the diameter at breast height (DBH; 1.3 m) of all trees over 5 m in height or 7 cm in diameter was recorded. The diameter of trees less than 7 cm DBH were measured in one 10 m² circular plot (1.72 m radius) at the plot center. Total basal area was estimated as the sum of the two measures. In each 1-m² sub-plot, the % cover of each non-vascular, graminoid or herbaceous plant species encountered was estimated, and the surface organic layers were characterized. The depth of the total surface organic layer and of L and F horizons were measured and the ratio of L:F was calculated as an indicator of faunal processing and mixing. Humus forms were classified according to Green et al. (1993) and Klinka et al. (1997). Activity levels of soil macrofauna activity in surface organic layers were assessed in the field based on visual assessment of the abundance of macrofaunal feces and casts present in the material. The abundance of feces was assigned a score between zero (none) and ten (abundant) (adapted from British Columbia Ministry of Forests and Range 2010).

Incorporation of faunal feces into the upper mineral soil (bioturbation) each sub-plot was also assessed using an adaptation of the stereomicroscope method of Frouz et al. (2009). Three 1-cm³ samples of the top 1 cm of mineral soil were collected and composited for each sub-plot. In the laboratory, each sample was spread evenly over a clear dish marked with a 1-cm² grid and examined under a dissecting microscope, recording occurrences of soil macrofauna, feces or castings. Each quarter of the dish was assessed separately, and assigned a score out of 25 based on the amount of feces or casting present. The scores for each dish and sub-plot were then averaged to provide an overall score for each site.

For sampling soils, a 30-cm-deep pit was dug within each sub-plot, and soil was described, including texture, structure and colour of all horizons. Soil samples (500 cm³) were collected in progressive depth intervals, from the base of the forest floor down to 30 cm in the mineral soil or PMM (0–5, 5–10, 10–20, 20–30 cm). In some sub-plots on reclaimed sites, the PMM did not extend the full 30 cm; in these cases the PMM and the underlying overburden were sampled separately. Samples were taken as intact blocks,

cutting a 5 × 10 × 10-cm block. Samples from the same depth in each of the three sub-plots were composited onsite. All samples were transported in a cooler to the lab, and then stored at 4 °C.

Root mass in each soil sample was determined using a combination of dry-sieving and hand-sorting. A sub-sample of approximately half of the total soil sample was air-dried for 48 h, and weighed. The sample was then dry-sieved through a 2-mm mesh. Material greater than 2 mm was hand-sorted into roots, woody debris (not from roots) and coarse fragments. These groups were then weighed and expressed as a proportion of overall sample weight.

Organic matter concentration of each air-dried, sieved sample was determined by loss on ignition. Other characteristics of the soils at these sites have been reported by Masse et al. (2017).

Data analysis

Analyses were performed in SAS 9.3 (SAS Institute, Cary NC). Significant differences among vegetation treatments in organic matter concentrations (total, upper 10 cm, lower 20 cm), and scores for macrofaunal activity, bioturbation and root abundance were tested through analysis of variance (PROC GLM). Simple linear regression (PROC REG) was used to assess relationships between organic matter concentrations with depth and root abundance, and between macrofaunal activity and L:F ratio and bioturbation. A significance threshold of $\alpha=0.05$ was used for all statistical analyses.

Results

Vegetation

Tree cover was similar (60–67%) at the Deciduous, Spruce and Natural sites, although stem density was higher at the Natural sites (Table 1). Natural sites (N=4) were dominated by aspen or jack pine, or a mix of the two species. Shrub cover was greatest at the Natural sites (30.5%), with the most common shrub species being *Ledum groenlandicum*, *Vaccinium myrtilloides*, *Vaccinium vitis-idaea* and *Viburnum edule*. Shrubs were also abundant at the Deciduous sites (15% cover); the most common species were *Rubus idaea* and *Rosa acicularis*. Grasses (*Festuca* sp., *Bromus inermis*) and the herb *Medicago sativa* dominated at the grassland sites (75% cover); forb cover was 18%. Bryophyte cover was 46% at the Spruce sites and 42% at Natural sites, primarily *Hylocomium splendens*, *Pleurozium schreberii*, and *Ptilium crista-castrense*. Detailed vegetation data for each site can be found in Table 1.

Surface organic layers

The surface organic layer at the Grass sites were all classified as lignomodors, which are associated with high levels of macrofaunal activity and little fungal activity (Green et al. 1993; Klinka et al. 1997). In contrast, forest floors in the Natural and Spruce sites were all classified as hemimors, which show little evidence of macrofaunal activity but high levels of fungal activity. Forest floors at three of the deciduous sites were classified as

Table 1 Composition of vegetation in each vegetation treatment

Vegetation treatment	Trees stems (ha)	Basal area (m ² /ha)	Tree cover (%)	Shrub cover (%)	Forb cover (%)	Grass cover (%)	Understory cover (%)	Non-vascular cover (%)
Deciduous	2650 ± 1015	15.95 ± 9.95	60.53 ± 11.73	14.68 ± 15.61	6.43 ± 4.49	20.78 ± 26.53	41.88 ± 20.86	3.50 ± 4.51
Grasslands	0	0	2.28 ± 4.48	1.58 ± 1.87	18.30 ± 19.73	75.25 ± 22.08	95.13 ± 8.62	4.25 ± 7.23
Spruce	1875 ± 655	8.17 ± 5.44	67.04 ± 12.30	5.10 ± 3.38	11.06 ± 11.74	0.94 ± 1.71	17.10 ± 11.41	46.00 ± 23.44
Natural	5275 ± 1438	14.47 ± 8.26	65.53 ± 21.52	30.50 ± 26.50	6.33 ± 2.91	0.50 ± 1.00	37.33 ± 26.80	42.10 ± 36.54

Each value is the mean of four sites (five for Spruce) ± one standard deviation

lignomodors, the other was characterized as a mormoder which is intermediate between hemimor and lignomoder.

The depth of the surface organic layer was greatest in the Natural sites and least at the Spruce sites (Table 2). The ratio of L:F depths (an indicator of litter processing and mixing) was highest in Grass sites, followed by Deciduous sites, and low in Spruce and Natural sites (Table 2). A similar pattern was evident for macrofaunal activity in the surface organic layer (Table 2). L:F ratio was positively correlated to macrofaunal activity in the surface organic layer ($r^2=0.537$; $p<0.0001$) (Fig. 1a).

Mineral soils/PMM

Average bioturbation scores in the mineral soil/PMM were highest in Deciduous sites (15.0), followed by Grass sites (10.4), and were low at the Spruce (3.0) and Natural (0.3) sites, which did not differ significantly from each other. Bioturbation activity was positively correlated with the abundance of signs of macrofaunal activity in the surface organic layer ($r^2=0.537$ $p=0.0004$; Fig. 1b). Bioturbation scores were also positively correlated with the concentration of organic matter in the upper 5 cm of mineral soil ($r^2=0.57$ $p=0.0004$), as well as the upper 10 cm ($r^2=0.50$ $p=0.0012$). Taken together, these relationships indicate higher levels of processing of surface organic matter by soil macrofauna in Grass and Deciduous sites than in Spruce and Natural sites.

The soils in all three vegetation treatments had significantly higher organic matter concentrations than the Natural soils (2.31%). SOM concentrations of soil organic matter in the upper 30 cm of mineral soil/PMM were significantly higher at the Deciduous sites (15.2%), than the Spruce (10.4%) and Grass sites (10.9%). SOM concentrations in the upper 10 cm of PMM at the Deciduous sites were significantly greater than those in the lower 20 cm (Fig. 2). A similar trend was apparent at Grass sites, although the difference was not significant. At Spruce and Natural sites SOM concentration in the upper 10 cm and the lower 20 cm of PMM material were not different. There was a significant, inverse relationship between organic matter concentration and depth in the PMM at the Deciduous sites ($r^2=0.563$; $p=0.01$; Fig. S1) and a similar trend at the Grass sites ($r^2=0.559$; $p=0.09$; Fig. S1), but no relationship with depth at the Spruce and Natural sites. Root mass was correlated with organic matter concentration only at Grass sites ($r^2=0.45$, $p=0.04$), although a similar trend is apparent at the Deciduous sites (Fig. 3).

The distribution pattern of SOM in each treatment is summarized in Fig. 4. Concentrations of SOM at 10–30-cm depth were similar (10–15%) in the three reclamation treatments. Organic matter concentrations were elevated in the upper 10 cm of soil at the Deciduous and Grass sites but not at the Spruce sites. Deciduous and Grass sites also had deeper

Table 2 Characteristics of the surface organic layers in each vegetation treatment

Vegetation treatment	Humus form	Depth (cm)	L:F Ratio	Macrofaunal activity score
Deciduous	Mormoder, Leptomoder	4.27 ± 1.67b	1.09 ± 0.59b	7.42 ± 3.34b
Grasslands	Leptomoder	3.66 ± 1.73b	1.96 ± 1.55a	9.17 ± 1.95a
Spruce	Hemimor	1.79 ± 0.48c	0.28 ± 0.22c	1.07 ± 1.03c
Natural	Hemimor	6.63 ± 3.40a	0.32 ± 0.19c	1.17 ± 1.03c

Each value is the mean of four sites (five for Spruce) ± one standard deviation. Within a column, different letters denote significant difference between means based on oneway ANOVA

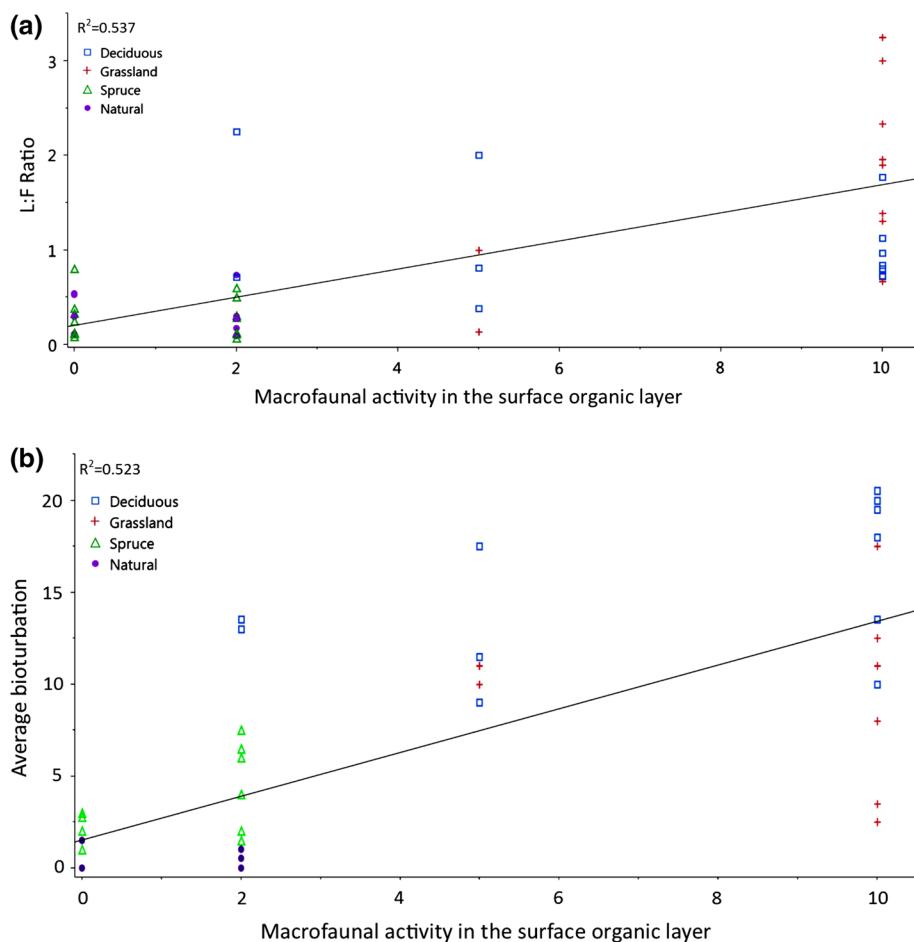
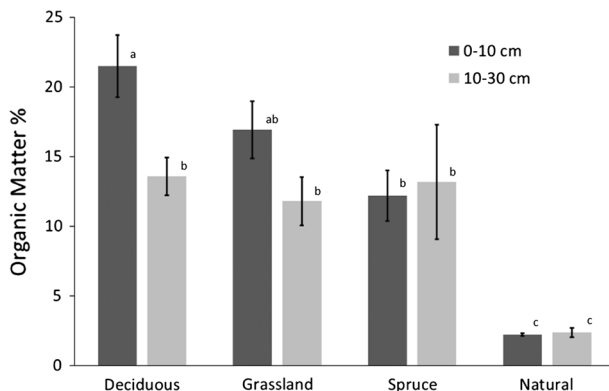


Fig. 1 **a** Relationship between macrofaunal activity score in surface organic layer and ratio of depths of L and F layers in each vegetation treatment at reclaimed sites and at natural sites. **b** Relationship between macrofaunal activity score in surface organic layer and bioturbation score in top 1 cm of peat-mineral mix in each vegetation treatment at reclaimed sites and mineral soil at natural sites

Fig. 2 Organic matter concentration (mean $\pm$ 1 standard deviation) of the upper 10 cm and the lower 20 cm of peat-mineral mix in each vegetation treatment at reclaimed sites and the mineral soil at natural sites. Each value is the mean of four sites (five for Spruce) $\pm$ one standard deviation



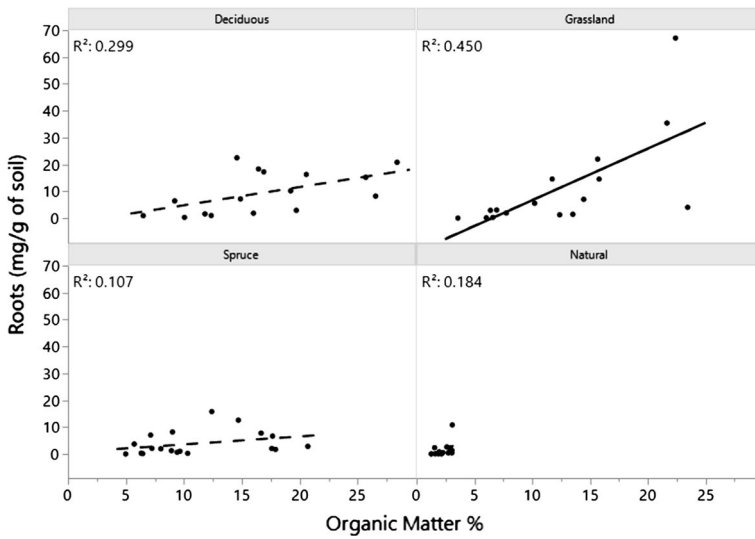


Fig. 3 Relationships between root mass and soil organic matter concentrations in the peat-mineral mix in each vegetation treatment at reclaimed sites and in the mineral soil at natural

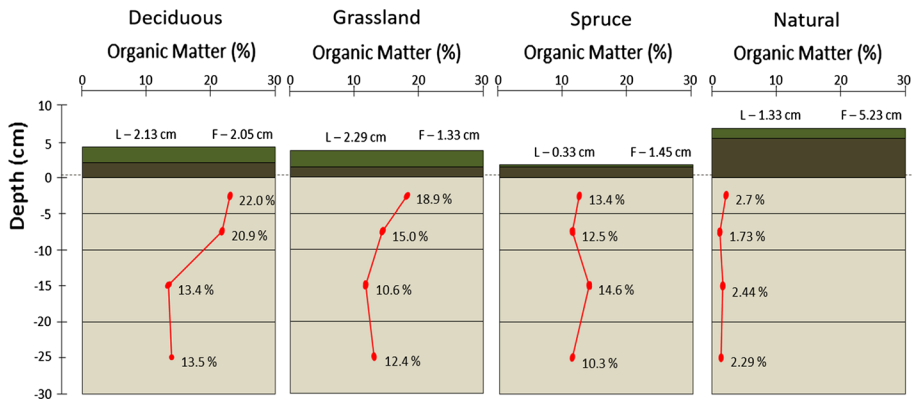


Fig. 4 Schematic representation of organic matter distribution with soil depth in each vegetation treatment at reclaimed sites and at natural sites

surface organic layers than the Spruce sites. At the Natural sites, organic matter concentrations were <3% and were slightly elevated in the uppermost 5 cm, but the surface organic layers were deeper than at the reclaimed sites.

Discussion

The substantially higher SOM concentrations in the upper 30 cm of the reclaimed soils compared to the natural soils indicate that much of the PMM coversoil remains 20–36 years after reclamation placement. The PMM is intended to provide a surface soil profile of high organic matter content like the forest floor (LFH horizon) of upland soils in the region, to serve as a growth medium for regenerating vegetation (Marty Yarmuch, personal communication). The positive relationship ($r^2=0.849$, $p<0.0001$) between SOM concentration and CEC found in the PMM coversoil in our study (Table S1, Figure S2) indicates that this organic matter continues to provide some of the positive benefits associated with SOM (e.g. CEC, moisture retention). Therefore, in contrast to many post-mining soils (Vindušková and Frouz 2013), reconstructed soils in the AOSR actually have much higher concentrations of organic matter (12–18%) in the upper 30 cm than do natural forest soils in the region (2.29%). With the exception of one spruce plot, average concentrations of SOM in all reclaimed sites studied were all above the 8% level, below which Farnden et al. (2013) detected reduced growth rates of trees in reconstructed oil sand soils. Accumulation of new soil organic matter is, therefore, not as critical for the development of the new ecosystem at sites reclaimed with a PMM coversoil as in post-mining soils which are impoverished in organic matter. However, evidence that new organic matter is accumulating in these reclaimed soils is an important indicator of soil development and ecosystem function.

There was no indication of new SOM accumulating in PMM planted with spruce, which is consistent with other observations that organic matter accumulates slowly and primarily on the surface of mine soils vegetated by spruce or other conifers (Vindušková and Frouz 2013). In common garden experiments, after several decades, *Picea* species often have as much accumulated soil C as deciduous species, but most of the C in spruce soils is in the forest floor rather than in the mineral soil (Vesterdal et al. 2008, 2013). Indeed, in mature (80–140 years old) stands in northwestern Alberta (Kishchuk 2004), forest floors under white spruce were deeper (average 13 cm) than those under aspen (9 cm). However, we found little accumulation of forest floor at the spruce sites. Sorenson et al. (2011) also reported thinner forest floors under spruce than aspen 16–33 years following reclamation in the AOSR. The slower development of a forest floor under spruce can be attributed to the long needle-retention time of *Picea glauca* in the boreal forest (approximately 9 years based on MAT of 0.7 °C; Reich et al. 2014). Smaller average mass of foliar litterfall has also been reported in spruce-dominated stands than aspen-dominated stands in boreal mixedwood forests of northern Alberta (Jerabkova et al. 2006). The apparent lack of new SOM accumulation at the spruce sites agrees with the conclusions of Laganier et al. (2010a) that afforestation with conifers has little effect on SOC accumulation, and that several decades may be necessary before there is a detectable increase in SOC in response to afforestation in the boreal zone.

The elevated concentrations of SOM in the upper PMM at the Grass sites supports our expectation based on earlier reports of faster soil C accumulation in mine soils under grass (Akala and Lal 2001) or pasture (Sperow 2006) than under forest canopy. In a study in the Italian Alps (Guidi et al. 2014), higher SOC concentrations in the top 10 cm of mineral soil were found in managed and abandoned grassland plots compared with nearby forests of spruce (*Picea abies*) and beech (*Fagus sylvatica*). The depth distribution of SOM, with elevated concentrations of SOM in the upper PMM relative to levels in the lower parts of the profile, are also consistent with the tendency for new organic matter to accumulate in the shallow portions of topsoil in prairie grasslands (O'Brien et al. 2010). We found

evidence of litter processing by soil fauna in the surface organic layer at the Grass sites, as has been reported in temperate prairies and grasslands (Curry 1993; Hunt et al. 1987; Vossbrink et al. 1979).

The elevated SOM levels in PMM at Grass sites also extended to the 10–20 cm depth, which, together with the correlation between SOM concentration and root abundance at Grass sites, suggests that some portion was derived from roots and rhizodeposits. A greater contribution from roots was expected at Grass sites, given the predominance of root biomass in grasslands (average root:shoot ratio of 3.7 compared to 0.32 in boreal forests; Jackson et al. 1996) and the abundance of root biomass in the upper soil (44% of total root biomass in top 10 cm in temperate grasslands; Jackson et al. 1996). Root litter is thought to be an important source of SOM due to its inherent recalcitrance, and its presence in the soil matrix where it can be protected in aggregates, and interact with clay minerals (Balesdent and Balabane 1996; Rasse et al. 2005). Wei et al. (2012) reported greater SOC content and deeper distribution of new SOC in grasslands compared to secondary forests and suggested that grasslands would be more effective for SOC sequestration and restoration of degraded soils in the Great Loess Plateau. Rhizodeposits (including exudates) are also thought to be important for SOM formation as these relatively labile carbon compounds stimulate growth of microbial biomass which leads to the production of stable organic matter (Cotrufo et al. 2015). Greater stability of SOC under grasses was indicated by the reduction in SOC in stable aggregates during natural forest succession on montane grasslands in Italy (Guidi et al. 2014) and the increase in SOC in the silt+clay soil size fraction when native forest was replaced by pasture in South America (Eclesia et al. 2012). Assessment of the relative amounts of SOC in the pools of various stabilities is needed to establish if grasses have the same effect on reclaimed mine sites in the boreal forest. Regardless of their potential for building soil organic matter, in the AOSR grasses are an option for restoration only on sites where a tree cover is not desired, as grasses can interfere with forest establishment in this region of the boreal forest (Liefers et al. 1993).

The elevated concentration of SOM in the upper PMM at the deciduous sites indicates accumulation of new SOM at the deciduous sites, and is consistent with other reports of faster soil C accumulation in mine soils under broadleaf species than under conifers (Frouz et al. 2009; Vindušková and Frouz 2013). In the meta-analysis of Laganier et al. (2010a), the average increase in SOC two to three decades following afforestation with broadleaf tree species was 25%, compared with 2% with conifers. Aspen forests (*Populus tremuloides* Michx.) have been found to store more SOC in mineral soil than do conifers and the SOC under aspen tends to be more stable (Boca and Van Miegroet 2017; Laganier et al. 2013, 2017).

In our study, there was clear evidence of macrofaunal litter processing activity in forest floors at the Deciduous sites. Likewise, Fons et al. (1998) commonly found a thin but well-developed Fz horizon (a forest-floor layer with high faunal activity) above an Fm horizon (a forest-floor layer with high fungal activity) in natural aspen forests (35–70 years old) in the western Canadian boreal forest. In eastern Canadian boreal forests, Laganier et al. (2009) found that aspen forest floors contained abundant macrofauna (i.e., earthworms, ants, snails, beetles), whereas microarthropods and Enchytraeidae dominated forest floors under black spruce. The epigeic earthworm, *Lumbricus rubellus*, was present exclusively in the forest floor of aspen stands (Laganier et al. 2010b). Although we did not survey soil fauna, one unidentified earthworm was encountered in an aspen forest floor during sampling. As suggested by Sorenson et al. (2011), examinations of soil fauna in regenerating forest in the AOSR are needed to understand the influence of vegetation treatment on forest floor and soil development on reclaimed mine sites in the AOSR.

Roots are probably also contributing to the formation of new SOM under the deciduous trees. *Populus* trees are known to have particularly thin fine roots, predominantly root orders 1 and 2 (Pregitzer et al. 2002), and thin roots tend to decompose more slowly than thicker roots (Goebel et al. 2011). *Populus tremuloides* has more roots in the mineral soil than does white spruce, the roots of which are concentrated in the forest floor (Strong and La Roi 1983). In a common garden experiment comparing 12 temperate tree species, *Populus tremuloides* had the highest specific root length (i.e. were thinnest), the shortest lifespan (mean of 153 days) and the highest concentrations of non-structural carbohydrates (starch and sugars), in addition to the fastest aboveground growth (McCormack et al. 2012). Greater rhizodeposition under aspen has been proposed as a mechanism underlying the greater stability of SOC under aspen (Laganiere et al. 2013, 2017) and its greater association with the silt and clay fraction of the soil relative to conifer species (Dobarco and Van Miegroet 2014; Boca and Van Miegroet 2017). All of these observations indicate that *Populus* trees may be particularly effective in building soil organic matter.

Leaching of DOM from the surface organic layer is probably also contributing to the elevated SOM concentrations in the upper topsoil in grassland and deciduous sites, but we have no measures from which to estimate this flux. Woldelessie et al. (2012) proposed that the high levels of SOC in aspen soils were in part due to rapid overwinter decomposition of aspen litter, which creates a pulse of DOC during snowmelt that is adsorbed on clay surfaces in the mineral soil. In grassland soils, Cotrufo et al. (2015) demonstrated that dissolved organic matter leached from litter early in decomposition are incorporated into microbial biomass at high rates, leading to formation of stable soil organic matter. At Spruce sites, the lack of increase in SOM in the surface PMM and the limited development of a forest floor indicates that leaching of DOM has not become an important flux at these sites.

Our findings of elevated organic matter concentrations in the upper 10 cm of PMM at the deciduous and grassland sites (but not spruce sites) are consistent with reports of higher rates of soil C accumulation in reclaimed soils under broadleaf species than under conifers (Frouz et al. 2009), and in soil under grasses and deciduous trees than beneath coniferous trees (Vinduřková and Frouz 2013). Likewise, our finding of fastest SOM accumulation and greatest bioturbation activity under deciduous trees supports the conclusion of Frouz et al. (2006, 2009) and Frouz (2008) that revegetating mine soils with tree species that promote soil macrofauna will hasten soil development. Laganiere et al. (2015) found that aspen abundance was linked with higher productivity, biomass and soil heterotrophic respiration, and suggested that aspen acts as a key driver of ecosystem C storage and fluxes in natural boreal forest ecosystems. Planting of aspen (alone or in mixtures with spruce) following reclamation would therefore be recommended to promote C sequestration into forest biomass and soil organic matter.

Conclusions

Twenty to 36 years following reclamation with peat-mineral mix, reclaimed soils had greater SOM concentrations than natural forest soils of similar age since stand-replacing disturbance. New soil organic matter appears to have accumulated in the upper 10 cm of soil at sites planted with deciduous trees, and at sites seeded to grasses, but not at sites planted with spruce. Given the mandate to restore oil sands sites with native boreal forest

species, afforestation with *Populus* is recommended to foster accumulation of new SOM and soil development.

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